Normal Distribution

①

Continuous r.v 'X' is normally distributed iff its p.d.f is

$$f(x) \Rightarrow f(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$-\infty < x < \infty$
$-\infty < \mu < \infty$
$\sigma > 0$

②

Normal distribution is the limiting form of the Binomial distribution, when 'n' is large neither 'p' nor 'q' is very small.

Properties of normal Distribution

1- The normal Distribution is a continuous distribution that ranges from $-\infty$ to ∞ ③

2- The total area under the normal curve is unity

3- The normal distribution is unimodal and its maximum ordinate at $x = \mu$

4- Normal Distribution is a symmetrical distribution

(i) Mean = Median = Mode

(ii) The lower and upper quartile are equidistant from its mean (4)

$$Q_3 - \mu = \mu - Q_1$$

(iii) All odd order moments about mean is zero.

5- If μ is the mean and σ is the S.D

(i) $p(\mu - \sigma < x < \mu + \sigma) = 0.6827$

(ii) $p(\mu - 2\sigma < x < \mu + 2\sigma) = 0.9545$

(iii) $p(\mu - 3\sigma < x < \mu + 3\sigma) = 0.9973$

In a normal distribution

$M.D = \frac{4}{5}\sigma$

$Q.D = \frac{2}{3}\sigma$

7- In normal distribution

$\beta_1 = 0$ and $\beta_2 = 3$

8- In a normal distribution the point of inflexion are $\mu - \sigma$ and $\mu + \sigma$

Question = 1

Show that the Area under the normal curve is unity.

Sol

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad -\infty < x < \infty$$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

L.H.S

$$\frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

put $\frac{x-\mu}{\sigma} = z$

$x = \mu + \sigma z$

$dx = \sigma dz$

$x = -\infty$ | $x = \infty$

$z = -\infty$ | $z = \infty$

$$\frac{1}{\sqrt{2\pi}.\cancel{\sigma}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} \cancel{\sigma}\, dz$$

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} dz$$

put $\frac{z^2}{2} = t \Rightarrow z^2 = 2t$

$\cancel{2}z\,dz = \cancel{2}dt$

$dz = \frac{dt}{z}$

$= \frac{dt}{\sqrt{2t}}$

$$\frac{1.2}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-\frac{1}{2}z^2} dz$$

$$\frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-t} \cdot \frac{dt}{\sqrt{2t}}$$

$$\Rightarrow \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-\frac{1}{2}} e^{-t} dt$$

$$\int_0^\infty x^{n-1} e^{-x} = \Gamma n \quad ③$$

$$\Rightarrow \frac{1}{\sqrt{\pi}} \int_0^\infty t^{\frac{1}{2}-1} e^{-t} dt$$

$$\Gamma n = (n-1)\Gamma(n-1)$$

$$(n-1)(n-2)\Gamma n-3$$

$$\Rightarrow \frac{1}{\sqrt{\pi}} \Gamma \tfrac{1}{2}$$

$$\Gamma \tfrac{1}{2} = \sqrt{\pi}$$

$$\Gamma 1 = 1$$

$$\Rightarrow \frac{1}{\cancel{\sqrt{\pi}}} \cdot \cancel{\sqrt{\pi}} = 1 \Rightarrow R.H.S$$

Question

Show that the $M.D = \frac{4}{5}\sigma$ in a normal distribution.

Sol By definition

$$M.D = E|x-\mu|$$

$$M.D. = \int_{-\infty}^{\infty} |x-\mu| f(x) dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} |x-\mu| e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

put $\frac{x-\mu}{\sigma} = z$

$x = \mu + \sigma z$

$dx = \sigma dz$

$x = -\infty \quad z = -\infty$

$x = \infty \quad z = \infty$

$$= \frac{1}{\sqrt{2\pi}\cancel{\sigma}} \int_{-\infty}^{\infty} |\sigma z| e^{-z^2/2} \cancel{\sigma} dz$$

$$= \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} |z| e^{-z^2/2} dz$$

$$= \frac{2\sigma}{\sqrt{2\pi}} \int_0^\infty z e^{-z^2/2} dz$$

$\frac{z^2}{2} = t$

$z^2 = 2t$

$\cancel{2}z dz = \cancel{2} dt$

$dz = \frac{dt}{z} = \frac{dt}{\sqrt{2t}}$

$$= \frac{2\sigma}{\sqrt{2\pi}} \int_0^\infty \cancel{\sqrt{2t}} e^{-t} \cdot \frac{1}{\cancel{\sqrt{2t}}} dt$$

$$M.D = \frac{\sqrt{2}\cdot\sqrt{2}\,\sigma}{\sqrt{2\pi}} \int_0^\infty t^{1-1} e^{-t}\, dt$$

$$= \frac{\sqrt{2}\,\sigma}{\sqrt{\pi}} \,\Gamma 1$$

$$M.D = \frac{4}{5}\sigma$$

Question

Show that in a normal Distribution

Mean = Median = Mode

Sol

$$E(x) = \int_{-\infty}^{\infty} x f(x)\, dx = 1$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} x\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\frac{x-\mu}{\sigma} = z$$

$$x = \mu + \sigma z$$

$$dx = \sigma dz$$

$$= \frac{1}{\sqrt{2\pi}\cancel{\sigma}} \int_{-\infty}^{\infty} (\mu + \sigma z)\, e^{-z^2/2}\, \cancel{\sigma}\, dz$$

$$= \frac{\mu}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} dz + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z\, e^{-z^2/2} dz$$

$$= \frac{\mu}{\cancel{\sqrt{2\pi}}} \cdot \cancel{\sqrt{2\pi}} + 0$$

$$E(x) = \mu$$

Median $\int_{-\infty}^{a} f(x)\, dx = \frac{1}{2}$

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$$\frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{a} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \frac{1}{2}$$

$$\frac{1}{\sqrt{2\pi}\cancel{\sigma}}\int_{-\infty}^{\frac{a-\mu}{\sigma}} e^{-z^2/2}\, \cancel{\sigma} dz = \frac{1}{2}$$

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\frac{a-\mu}{\sigma}} e^{-z^2/2}\, dz = \frac{1}{2} \quad \text{—— (1)}$$

$\frac{x-\mu}{\sigma} = z$

$x = \mu + \sigma z$

$dx = \sigma dz$

$x = -\infty \quad z = -\infty$

$x = a \quad z = \frac{a-\mu}{\sigma}$

we know that for symmetry standard normal Distribution

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{0} e^{-z^2/2} dz = \frac{1}{\sqrt{2\pi}}\int_{0}^{\infty} e^{-z^2/2} \Rightarrow \frac{1}{2} \quad \text{—— (2)}$$

Comparing (1) and (2)

$$\frac{a-\mu}{\sigma} = 0$$

$$\underline{a = \mu}$$

For Mode

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Diff. w.r. to x.

$$f'(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \cdot \left(-\frac{2}{2}\left(\frac{x-\mu}{\sigma}\right)\left(\frac{1}{\sigma}\right)\right)$$

$$f'(x) = -\frac{1}{\sqrt{2\pi}\sigma^3} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}(x-\mu)$$

$$f''(x) = -\frac{1}{\sqrt{2\pi}\sigma^3}\left[(x-\mu)\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}\left(-\frac{\cancel{2}}{\cancel{2}}\left(\frac{x-\mu}{\sigma}\right)\left(\frac{1}{\sigma}\right)\right) + e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}\right]$$

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$$f''(x) = -\frac{1}{\sqrt{2\pi}\sigma^3} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}\left[-(x-\mu)\left(\frac{x-\mu}{\sigma^2}\right)+1\right]$$

$$= -\frac{1}{\sqrt{2\pi}\sigma^3} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}\left[1-\left(\frac{x-\mu}{\sigma}\right)^2\right]$$

By putting $x = \mu$

$$= -\frac{1}{\sqrt{2\pi}\sigma^3} e^{0}(1-0)$$

$$f''(\mu) = -\frac{1}{\sqrt{2\pi}\sigma^3}$$

$$f''(\mu) < 0$$

put $f'(x) = 0$

$$-\frac{1}{\sqrt{2\pi}\sigma^3} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}(x-\mu) = 0$$

$$x - \mu = 0$$

$$\boxed{x = \mu}$$

Question

Show that the m.g.f of Normal Distribution is $M_0(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$

Sol

By definition of m.g.f

$$M_0(t) = E(e^{tx})$$

$$= \int_{-\infty}^{\infty} e^{tx} f(x)\, dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} e^{tx} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

put $\frac{x-\mu}{\sigma} = z$

$x = \mu + \sigma z$

$dx = \sigma dz$

$x = -\infty$, $z = -\infty$ | $x = \infty$, $z = \infty$

$$M_0(t) = \frac{1}{\sqrt{2\pi}\,\sigma} \int_{-\infty}^{\infty} e^{t(\mu+\sigma z)}\, e^{-z^2/2}\, \sigma\, dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\mu t} \cdot e^{t\sigma z} \cdot e^{-z^2/2}\, dz$$

$$= \frac{e^{\mu t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\sigma z t - z^2/2}\, dz$$

$$= \frac{e^{\mu t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z^2 - 2\sigma z t)}\, dz$$

$$= \frac{e^{\mu t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z^2 - 2\sigma z t + \sigma^2 t^2 - \sigma^2 t^2)}\, dz$$

$$= \frac{e^{\mu t} \cdot e^{\frac{1}{2}\sigma^2 t^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-\sigma t)^2}\, dz$$

$$= \frac{e^{\mu t + \frac{1}{2}\sigma^2 t^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-\sigma t)^2}\, dz$$

$z - \sigma t = w$ $\quad$ $z = -\infty$, $w = -\infty$ $\quad$ $z = \infty$, $w = \infty$

$dz = dw$

$$= \frac{e^{\mu t + \frac{1}{2}\sigma^2 t^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{w^2}{2}}\, dw$$

$$= \frac{2\left(e^{\mu t + \frac{1}{2}\sigma^2 t^2}\right)}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-w^2/2}\, dw$$

$\frac{w^2}{2} = t$

$w^2 = 2t$

$\cancel{2}w\,dw = \cancel{2}dt$

$dw = \frac{dt}{w}$

$= \frac{dt}{\sqrt{2t}}$

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$$M_0(t) = \frac{2\left(e^{\mu t + \frac{1}{2}\sigma^2 t^2}\right)}{\sqrt{2\pi}} \int_0^\infty e^{-t} \frac{dt}{\sqrt{2t}}$$

$$= \frac{e^{\mu t + \frac{1}{2}\sigma^2 t^2}}{\sqrt{\pi}} \int_0^\infty t^{-\frac{1}{2}} e^{-t}\, dt$$

$$= \frac{e^{\mu t + \frac{1}{2}\sigma^2 t^2}}{\sqrt{\pi}} \int_0^\infty t^{\frac{1}{2}-1} e^{-t}\, dt$$

$$= \frac{e^{\mu t + \frac{1}{2}\sigma^2 t^2}}{\sqrt{\pi}} \cdot \Gamma\left(\tfrac{1}{2}\right)$$

$$M_0(t) = \frac{e^{\mu t + \frac{1}{2}\sigma^2 t^2}}{\cancel{\sqrt{\pi}}} \cdot \cancel{\sqrt{\pi}}$$

$$\underline{M_0(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}}$$

<u>Question</u> prove That the normal curve has point of inflexion which are equidistant from mean.

<u>Sol</u> We have.

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{——— (i)}$$

$$f'(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}\left[-\frac{1}{\cancel{2}} \cdot \cancel{2}\left(\frac{x-\mu}{\sigma}\right)\left(\frac{1}{\sigma}\right)\right]$$

$$= -\frac{1}{\sqrt{2\pi}\sigma^3} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}(x-\mu)$$

Diff. again w. r. to (x)

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$$f''(x) = -\frac{1}{\sqrt{2\pi}\sigma^3}\left[(x-\mu)\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \cdot -\frac{2}{2}\left(\frac{x-\mu}{\sigma}\right)\left(\frac{1}{\sigma}\right) + e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}(1)\right]$$

$$= -\frac{1}{\sqrt{2\pi}\sigma^3}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}\left[-\frac{(x-\mu)^2}{\sigma^2} + 1\right]$$

$$f''(x) = 0$$

$$\frac{1}{\sqrt{2\pi}\sigma^3} > 0 \quad \text{and} \quad e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} > 0$$

$$1 - \frac{(x-\mu)^2}{\sigma^2} = 0$$

$$\left(\frac{x-\mu}{\sigma}\right)^2 = 1$$

$$\frac{x-\mu}{\sigma} = \pm 1$$

$$x - \mu = \pm\sigma$$

$$x - \mu = \sigma \quad\Big|\quad x - \mu = -\sigma$$

$$\underline{x = \mu + \sigma} \quad\Big|\quad \underline{x = \mu - \sigma}$$

put these values in eq $\Rightarrow$ $x = \mu + \sigma$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma}\, e^{-\frac{1}{2}\left(\frac{\mu+\sigma-\mu}{\sigma}\right)^2} \Rightarrow \frac{1}{\sigma\sqrt{2\pi e}}$$

put $x = \mu - \sigma$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma}\, e^{-\frac{1}{2}\left(\frac{\mu-\sigma-\mu}{\sigma}\right)^2} \Rightarrow \frac{1}{\sigma\sqrt{2\pi e}}$$

Hence point of inflexion of normal curve are

$$\left(\mu - \sigma, \frac{1}{\sigma\sqrt{2\pi e}}\right) \text{and} \left(\mu + \sigma, \frac{1}{\sigma\sqrt{2\pi e}}\right)$$

Question

Find the mean m.g.f. use it to find β_1 and β_2.

Sol By definition

$$M_{(x-\mu)}t \Rightarrow E\left(e^{t(x-\mu)}\right)$$

$$\Rightarrow E\left(e^{tx} \cdot e^{-\mu t}\right)$$

$$\Rightarrow e^{-\mu t} E\left(e^{tx}\right)$$

$$\Rightarrow e^{-\mu t} \cdot e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

$$M_{(x-\mu)}t \Rightarrow e^{\frac{1}{2}\sigma^2 t^2}$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$= 1 + \frac{1}{2}\sigma^2 t^2 + \frac{1}{2!}\left(\frac{1}{2}\sigma^2 t^2\right)^2 + \frac{1}{3!}\left(\frac{1}{2}\sigma^2 t^2\right)^3 + \frac{1}{4!}\left(\frac{1}{2}\sigma^2 t^2\right)^4 + \cdots$$

$$M_{(x-\mu)}t = 1 + \frac{1}{2}\sigma^2 t^2 + \frac{1}{2}\cdot\frac{1}{4}\sigma^4 t^4 + \frac{1}{6}\left(\frac{1}{8}\sigma^6 t^6\right) + \cdots$$

$$= 1 + \frac{1}{2!}\sigma^2 t^2 + \frac{3\sigma^4 t^4}{4!} + \frac{\sigma^6 t^6}{48} + \cdots$$

As there is no odd powers, so all odd order moments are zero

$$\mu_{2n+1} = 0$$

For even order moments compare the coefficients of $\frac{t^r}{r!}$

$$\mu_2 = \sigma^2 \qquad \mu_4 = 3\sigma^4$$

$$\beta_1 = \frac{(\mu_3)^2}{(\mu_2)^3} \Rightarrow \frac{0}{(\sigma^2)^3} = 0$$

$$\beta_2 = \frac{\mu_4}{(\mu_2)^2} \Rightarrow \frac{3\sigma^4}{(\sigma^2)^2} \Rightarrow 3$$

Question

Calculate Cumulant generating function, also calculate Moments, γ_1 and γ_2

Sol We know that m.g.f

$$M_0(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

Taking log both sides

$$\ln(M_0(t)) = K(t) = \mu t + \frac{1}{2}\sigma^2 t^2$$

By comparing the coefficients of $\frac{t^r}{r!}$

$K_1 = \mu_1' = \mu$ | $K_3 = 0$

$K_2 = \sigma^2$ | $K_4 = 0$

$\mu_1 = 0$ $\quad$ $\mu_2 = K_2 = \sigma^2$

$\mu_3 = K_3 = 0$ $\quad$ $\mu_4 = K_4 + 3K_2^2$

$$= 0 + 3(\sigma^2)^2$$

$$= 3\sigma^4$$

$$\gamma_1 = \frac{K_3}{(K_2)^{3/2}} = \frac{0}{(\sigma^2)^{3/2}} = 0 \text{ (symmetrical)}$$

$$\gamma_2 = \frac{K_4}{(K_2)^2} = \frac{0}{(\sigma^2)^2} = 0 \text{ (Mesokurtic)}$$

Q) Show that normal distribution is the limiting form of the Binomial distribution:

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Sol: Normal distribution is the limiting form of the binomial distribution when

i) n, the no. of trials are indefinitely large i.e $n \to \infty$

ii) Neither 'p' nor 'q' is very small

the probability mass function of binomial distribution is given by

$$p(X=x) = \binom{n}{x} p^x q^{n-x} \quad \text{——(i)} \quad x = 0, 1, 2, \dots, n$$

with mean $E(x) = np$

and variance $V(x) = npq$

let us introduce the standard binomial variate.

$$z = \frac{x - E(x)}{\sqrt{V(x)}}$$

$$z = \frac{x - np}{\sqrt{npq}} \quad \text{——(ii)}$$

Now by definition

$$M_Z(t) = E(e^{tZ}) = E\left[e^{t(x-np)/\sqrt{npq}}\right]$$

$$= e^{-npt/\sqrt{npq}}\, E\left(e^{tx/\sqrt{npq}}\right)$$

$$= e^{-npt/\sqrt{npq}} \sum_{x=0}^{n} e^{tx/\sqrt{npq}}\, p(x)$$

$$= e^{-npt/\sqrt{npq}} \sum_{x=0}^{n} e^{tx/\sqrt{npq}} \binom{n}{x} p^x q^{n-x}$$

$$= e^{-npt/\sqrt{npq}} \sum_{x=0}^{n} \binom{n}{x} \left(p e^{t/\sqrt{npq}}\right)^x q^{n-x}$$

$$= e^{-npt/\sqrt{npq}} \left(q + p e^{t/\sqrt{npq}}\right)^n$$

$$= \left[e^{-pt/\sqrt{npq}} \left(q + p e^{t/\sqrt{npq}}\right)\right]^n$$

$$= \left(q e^{-pt/\sqrt{npq}} + p e^{qt/\sqrt{npq}}\right)^n$$

$E(e^{tx})$

$= \Sigma e^{tx} p(x)$

$= \Sigma e^{tx} \binom{n}{x} p^x q^{n-x}$

$(q+pe^t)^n$

$p e^{t/\sqrt{npq}}\, e^{-pt/\sqrt{npq}}$

$p e^{\frac{t}{\sqrt{npq}} - \frac{pt}{\sqrt{npq}}}$

$\frac{t}{\sqrt{npq}}(1-p)$

$p e^{qt/\sqrt{npq}}$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$= \left[q\left\{1 + \frac{-pt}{\sqrt{npq}} + \frac{1}{2!}\left(\frac{-pt}{\sqrt{npq}}\right)^2 + \cdots\right\} + p\left\{1 + \frac{qt}{\sqrt{npq}} + \frac{1}{2!}\left(\frac{qt}{\sqrt{npq}}\right)^2 + \cdots\right\}\right]^n$$

$$M_Z(t) = \left[q - \frac{qpt}{\sqrt{npq}} + \frac{q}{2!}\frac{p^2t^2}{npq} + \cdots + p + \frac{qpt}{\sqrt{npq}} + \frac{p}{2!}\frac{q^2t^2}{npq} + \cdots\right]^n$$

$$= \left[(q+p) + (p+q)\frac{1}{n}\frac{t^2}{2!} + \frac{1}{3!}(\;) + \cdots\right]^n$$

$$= \left[1 + \frac{1}{n}\frac{t^2}{2!} + \frac{1}{3!}(\;) + \cdots\right]^n$$

taking ln both sides

$$\ln M_Z(t) = \ln\left(1 + \frac{1}{n}\frac{t^2}{2!} + \frac{1}{3!}(\text{———}) + \text{—}\ldots\right)^n$$

$$= n \ln\left(1 + \frac{1}{n}\frac{t^2}{2!} + \frac{1}{3!}(\quad) + \text{——}\right)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - - - \text{—}$$

$$= n\left[\left(\frac{1}{n}\frac{t^2}{2!} + \frac{1}{3!}(\quad) + \cdots\right) - \left(\frac{1}{n}\frac{t^2}{2!} + \frac{1}{3!}(\quad) + \text{—}\right)^2 + \text{—}\ldots\right]$$

As limit $n \to \infty$

$$\ln M_Z(t) = \frac{t^2}{2!} + 0$$

$$\ln M(t)^t = \frac{t^2}{2}$$

$$M_Z(t) = e^{t^2/2}$$

which is the m.g.f of the standard normal Distribution!